VAIMAANIK 2026 | INTERCOLLEGIATE TECHNICAL COMPETITION

OFFICIAL COMPETITION RULEBOOK

1. Event Objective

The Water Rocket Challenge evaluates engineering design and flight performance. Each team will design, build and launch a water-powered rocket. Each team is allowed two official launch attempts; only its higher-scoring attempt will count toward the final ranking.

2. Team Eligibility and Registration

1. Each team shall have exactly 2 members.
2. All participants must carry valid college identification.
3. Each participant may register with only one team.
4. Teams must bring their own rocket and complete registration before the event.
5. Teams must report at the time announced by the organizers.

3. Technical Specifications

Parameter	Requirement
Pressure vessel	Carbonated soft-drink PET pressure vessel, up to 2.5 L, subject to inspection and verified working-pressure suitability
Propulsion	Water and compressed air only
Launch pressure	Maximum pressure to be set by organizers after verifying the vessel and complete launcher assembly; no universal PSI limit is assumed safe
Nose cone	Lightweight, non-metallic, non-sharp
Fins	Allowed; securely attached
Recovery system	Not required; no recovery points
Launch pad / launcher	Teams must bring their own; none will be provided by organizers
Air pump / compressor	Teams must bring their own; none will be provided by organizers
Rocket stages	Single-stage and two-stage water rockets are permitted; maximum TWO stages, subject to safety inspection.

A maximum of TWO rocket stages is permitted. Both single-stage and two-stage designs are eligible. Any stage-separation mechanism and all pressurized components must pass inspection; unsafe or uncontrolled separation is prohibited. Each stage may use a maximum 2.5 L carbonated

soft-drink PET pressure vessel (5 L combined maximum for two stages). Both pressure chambers and the stage-separation mechanism require safety inspection.

Rockets must be fabricated by participating teams. Commercial ready-made rockets, hazardous materials, combustion systems, sharp metal tips and damaged pressure vessels are prohibited.

4. Launching Equipment and Inspection

6. The organizers will NOT provide a launch pad, launcher, pump or compressor.

7. Each team is responsible for transporting, assembling and operating its own approved equipment.

8. Launchers must be stable, securely anchored and equipped with a safe release arrangement that permits operation from behind the designated safety boundary.

9. Pressure vessels, fittings, hoses, release mechanisms and launcher supports must pass inspection before pressurization.

10. Teams must follow the organizer-approved pressure limit, launch direction and launch angle.

11. Only authorized persons may pressurize or release a rocket; the launch official must give clearance first.

12. Unsafe equipment will not be allowed to launch.

5. Competition Format — Two Launch Attempts (Best Score Counts)

13. Each team receives a maximum of TWO official launch attempts.

14. Flight time, horizontal distance and flight-path straightness are recorded and scored separately for each attempt. The higher TOTAL score of the two attempts is the team's final score. Scores from separate attempts must not be combined.

15. Flight time starts when the rocket leaves the launcher and ends at first ground contact.

16. Horizontal distance is measured along the ground from the launch point to the first ground-contact point.

17. Judges will also assess deviation from the designated flight direction and overall stability. A normal curved ballistic trajectory is not itself a straightness fault.

18. A replacement attempt may be granted only for an organizer-confirmed equipment or measurement failure. An additional tie-breaker launch may be authorized under Section 7.

6. Scoring System

Criterion	Points
Horizontal distance	10 points for each COMPLETE 5 m
Flight time	10 points for each COMPLETE 1 s
Flight-path straightness	0–20 points, awarded by judges

Distance points = 10 × floor(distance in metres ÷ 5).

Time points = 10 × floor(flight time in seconds).

Attempt score = Distance points + Time points + Straightness points. Final team score = the HIGHER attempt score (Launch 1 or Launch 2).

Only complete 5-metre intervals and complete seconds earn points. There is no overall maximum score; straightness is capped at 20 points per attempt. If both attempts have equal total points, the attempt with the greater actual measured horizontal distance is used for the team's recorded result.

6.1 Flight-path straightness rubric

Flight quality	Judges' points
Excellent – stable, closely aligned with intended direction	16–20
Good – minor directional deviation	11–15
Average – noticeable deviation	6–10
Poor – highly unstable or substantially off course	0–5

7. Tie-Breaking Criteria

19. Higher actual measured horizontal distance (not just distance points).
20. If still tied, longer actual measured flight time (not just time points).
21. If still tied, higher flight-path straightness score.
22. If still tied, judges may authorize one additional launch for the tied teams under the same conditions.

8. Safety Regulations

23. Only water and compressed air are permitted; chemicals, flames, explosives and other propellants are prohibited.
24. Spectators and participants must remain outside the marked launch and landing exclusion zones.
25. No one may stand in front of the launcher or approach a pressurized rocket.
26. Pressure must be monitored with a suitable gauge and must never exceed the approved limit.
27. Never pressurize a cracked, damaged or visibly deformed bottle.
28. Launches must be suspended in unsafe weather or if the landing area is not clear.
29. Retrieval is permitted only after the safety officer declares the area safe.
30. Judges and safety officials may stop any launch deemed unsafe.

9. General Rules

31. No recovery, parachute, design or innovation points will be awarded. Each team is permitted up to two official launches; the best total attempt score counts.
32. Teams must not interfere with other participants or equipment.
33. Rockets and launchers must remain compliant after inspection; changes require reinspection.
34. Failure to report when called may result in forfeiture of an attempt.
35. Judges' scoring and safety decisions are final.

36. Organizers may revise operational requirements for safety or fairness and will communicate changes to all teams.

10. Disqualification

- Use of prohibited propellants or hazardous construction.
- Exceeding the approved pressure limit or launching without clearance.
- Failure of mandatory rocket or launcher inspection.
- Unsafe conduct, interference, or deliberate manipulation of measurements.
- Repeated refusal to follow safety instructions.

11. Results and Awards

Teams are ranked by the higher total score achieved in either of their two official launch attempts. First, second and third positions go to the three highest-scoring eligible teams, subject to the tie-break rules.

Event Coordinators

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